

AP-E12 Fixed-Degree Complete-Minor Graph Density: Exact Strong Quasiconvexity, an Endpoint Dipole No-Go, and the Fail-Closed Relaxed-Minimizer Regularity Branch

Route-F derivation ledger

20 July 2026

Abstract

AP-E11 left one precise continuum gate: strong smooth density, at fixed degree, for maps whose first, second, and third derivative minors all belong to L^2 . AP-E12 audits that endpoint instead of assuming it. Two useful model-specific theorems are proved. First, graph-norm convergence of sphere-valued maps automatically preserves degree, so “fixed degree” adds no second repair once target-valued graph density is known. Second, the complete-minor density has an exact strong-quasiconvexity identity and uniform Legendre–Hadamard constant one. A scale-sharp calculation also excludes the direct Malý two-hole/dipole mechanism: finite L^2 second- minor energy requires $\alpha > 1/2$, while a non-vanishing forced filling cost requires $\alpha \leq 1/2$, including the logarithmic borderline.

These results do not prove the endpoint density theorem. The classical Cartesian approximation theorem loses exponent, and area-approximation results control all minors only in L^1 . The fallback partial-regularity theory also does not apply: the present density has $(p, q) = (2, 6)$ growth in dimension three, outside the recent relaxed-quasiconvex range $q < 3$, and it has an additional $\mathbb{S}^3$ constraint and fixed-degree relaxation. A finite-grid concentration and perturbed-start card passes 9/9 and gives a very strong same-basin proxy after quotienting translations and target $SO(3)$, but is not a continuum isolation theorem. Therefore endpoint density is neither proved nor disproved; partial regularity, classicality, continuum isolation, the Riemann Hessian, and determinant variation all remain false in the authorization ledger.

Contents

1	The exact endpoint statement	2
2	Exact algebra of the AP-E11 continuum density	2
3	What the Cartesian approximation theorems do and do not give	3
4	A scale-sharp dipole audit	3
5	Finite energy does not imply continuity	4
6	The relaxed-minimizer fallback and its exact obstruction	4
7	Numerical concentration and isolation proxy	5
8	Decision and next theorem	6

1 The exact endpoint statement

Let $\Omega \subset \mathbb{R}^3$ be a bounded Lipschitz domain, with constant vacuum trace $e_0 = (1, 0, 0, 0)$, and let $M_k(F)$ denote the vector of all $k \times k$ minors of $F \in \mathbb{R}^{4 \times 3}$. Define the complete-minor graph class

$$\mathcal{G}_B^2(\Omega, \mathbb{S}^3) = \left\{ u \in W^{1,2}(\Omega; \mathbb{S}^3) \mid \begin{array}{l} \text{tr } u = e_0, \text{ deg } u = B, \\ M_2(Du), M_3(Du) \text{ are distributional minors} \end{array} \right\}. \quad (1)$$

The strong graph distance is

$$d_G(u, v) = \|u - v\|_{L^2} + \|Du - Dv\|_{L^2} + \|M_2(Du) - M_2(Dv)\|_{L^2} + \|M_3(Du) - M_3(Dv)\|_{L^2}. \quad (2)$$

For sphere-valued maps the degree is

$$B(u) = \frac{1}{2\pi^2} \int_{\Omega} \det[u, \partial_1 u, \partial_2 u, \partial_3 u] \, dx. \quad (3)$$

It is integrable because $u \in L^\infty$ and $M_3(Du) \in L^2$.

The missing AP-E11 assertion is

$$\boxed{C_{e_0}^\infty(\bar{\Omega}; \mathbb{S}^3) \cap \{\text{deg} = B\}}^{d_G} = \mathcal{G}_B^2(\Omega, \mathbb{S}^3). \quad (D_{\text{end}}) \quad (4)$$

Ordinary $W^{1,2}$ density is available because the relevant obstruction is $\pi_2(\mathbb{S}^3) = 0$ [1]; it does not imply (4). In particular, strong L^2 convergence of Du_j gives only L^1 control of quadratic products, not strong L^2 control of $M_2(Du_j)$, and gives no endpoint control of cubic products.

Proposition 1.1 (Degree is automatically closed in the graph norm). *Suppose $u_j, u \in W^{1,2}(\Omega; \mathbb{S}^3)$ have the vacuum trace and $d_G(u_j, u) \rightarrow 0$. Then $B(u_j) \rightarrow B(u)$. If the u_j are smooth, their integer degrees therefore equal $B(u)$ for all sufficiently large j .*

Proof. Write $J_3(Du)$ for the signed four-vector dual to $M_3(Du)$. Since $\|u_j\|_\infty = \|u\|_\infty = 1$,

$$\begin{aligned} 2\pi^2 |B(u_j) - B(u)| &\leq \int_{\Omega} |(u_j - u) \cdot J_3(Du_j)| + |u \cdot [J_3(Du_j) - J_3(Du)]| \, dx \\ &\leq \|u_j - u\|_2 \|M_3(Du_j)\|_2 + |\Omega|^{1/2} \|M_3(Du_j) - M_3(Du)\|_2. \end{aligned} \quad (5)$$

The right side tends to zero, because the second graph component also bounds $\|M_3(Du_j)\|_2$. Integers converging to the integer $B(u)$ are eventually equal to it. $\square$

Thus the real missing theorem is target-valued endpoint graph density. A separate degree-changing patch must not be inserted; if graph convergence is proved, degree follows from (5).

2 Exact algebra of the AP-E11 continuum density

For $R, K > 0$ put

$$W(F) = \frac{1}{2} (|F|^2 + R|M_2(F)|^2 + K|M_3(F)|^2). \quad (6)$$

If s_1, s_2, s_3 are the singular values of F , Cauchy–Binet gives

$$2W(F) = \sum_i s_i^2 + R \sum_{i < j} s_i^2 s_j^2 + K s_1^2 s_2^2 s_3^2. \quad (7)$$

This is convex in the complete-minor vector $(F, M_2(F), M_3(F))$, but not in F itself.

Theorem 2.1 (Uniform rank-one symbol). *For every $F \in \mathbb{R}^{4 \times 3}$ and every rank-one $H = a \otimes \xi$,*

$$D^2W(F)[H, H] = |H|^2 + R|DM_2(F)[H]|^2 + K|DM_3(F)[H]|^2 \geq |a|^2 |\xi|^2. \quad (8)$$

Hence the Legendre–Hadamard constant is one, independently of F .

Proof. Every minor of $F + t(a \otimes \xi)$ is affine in t : a determinant cannot contain two columns from the same rank-one update. Taking two derivatives of the squared Euclidean norms in (6) gives (8). Finally $|a \otimes \xi|^2 = |a|^2 |\xi|^2$. $\square$

Theorem 2.2 (Exact strong quasiconvexity identity). *Let U be a periodic cell, F constant, and $\varphi \in W_{\text{per}}^{1,\infty}(U; \mathbb{R}^4)$ have zero mean. Then*

$$\begin{aligned} & \int_U \{W(F + D\varphi) - W(F)\} \, dx \\ &= \frac{1}{2} \int_U \{|D\varphi|^2 + R|M_2(F + D\varphi) - M_2(F)|^2 + K|M_3(F + D\varphi) - M_3(F)|^2\} \, dx. \end{aligned} \quad (9)$$

The same formula holds for zero-boundary variations on a bounded domain.

Proof. Every first, second, and third minor is a null Lagrangian. Periodicity gives

$$\frac{1}{|U|} \int_U D\varphi = 0, \quad \frac{1}{|U|} \int_U M_k(F + D\varphi) = M_k(F), \quad k = 2, 3. \quad (10)$$

Expand each squared norm in (6). All cross terms integrate to zero by (10); the remaining squares are precisely (9). $\square$

Equation (9) is stronger than a generic quasiconvexity estimate and is the main positive analytic structure discovered in AP-E12. It controls complete-minor excess whenever one can construct a competitor with the correct trace. It does not itself construct the trace-preserving, target-valued endpoint recovery sequence needed in (4).

3 What the Cartesian approximation theorems do and do not give

Malý's approximation theorem for Cartesian maps gives strong convergence of all minors in exponents $q_k < p_k$; it explicitly does not claim a bounded approximating sequence at the endpoint [2]. Mucci's slicing characterization treats strong area approximation, equivalently $W^{1,1}$ plus L^1 convergence of all minors [3]. The Cartesian-current framework supplies the correct closure language for maps into spheres [4]. None of these statements is the L^2 endpoint (4).

There is a useful conditional implication. If the selected map belongs to the appropriate target-valued Cartesian closure, its complete-minor vector has local $L^{2+\delta}$ control for some $\delta > 0$, and the boundary/ $\mathbb{S}^3$ projection step can be performed without losing that control, then Malý's exponent-loss theorem may be applied with output exponent two. This reduces one route to the higher-integrability estimate

$$(Du, M_2(Du), M_3(Du)) \in L_{\text{loc}}^{2+\delta}. \quad (H_\delta) \quad (11)$$

The target projection hypothesis is essential: ambient smooth approximation alone does not keep values on $\mathbb{S}^3$.

4 A scale-sharp dipole audit

The natural counterexample attempt is a shrinking two-dimensional Cartesian dipole extended along a third coordinate. In a target chart write

$$u(r, \theta, z) = a(r)\gamma(\theta), \quad (12)$$

where γ carries the nontrivial two-hole filling datum. Up to harmless angular constants,

$$\int |Du|^2 \sim \int_0^1 a'(r)^2 r + \frac{a(r)^2}{r} \, dr, \quad (13)$$

$$\int |M_2(Du)|^2 \sim \int_0^1 \frac{a(r)^2 a'(r)^2}{r} \, dr. \quad (14)$$

Lemma 4.1 (No power-law overlap). *For $a(r) = r^\alpha$, finite Dirichlet energy requires $\alpha > 0$, while finite L^2 second-minor energy requires $\alpha > 1/2$. Any smoothing that fills a fixed nonzero signed target area on a disk of radius r obeys*

$$\int_{D_r} |\det Dv|^2 \, dx \geq \frac{1}{\pi r^2} \left| \int_{D_r} \det Dv \, dx \right|^2 \gtrsim \frac{a(r)^4}{r^2} = r^{4\alpha-2}. \quad (15)$$

Thus a non-vanishing or divergent endpoint filling cost requires $\alpha \leq 1/2$, which has no overlap with (14).

Proof. Substitution into (13) gives an integrand proportional to $r^{2\alpha-1}$; substitution into (14) gives $r^{4\alpha-3}$. The stated integrability thresholds follow. Inequality (15) is Cauchy–Schwarz on D_r , and the signed filling area scales as $a(r)^2$. $\square$

The logarithmic borderline also fails to open a gap. For

$$a(r) = r^{1/2} [\log(e/r)]^{-\beta}, \quad (16)$$

the second-minor integral is finite exactly when $\beta > 1/4$, whereas the lower bound in (15) is $[\log(e/r)]^{-4\beta} \rightarrow 0$ for every $\beta > 0$. Consequently:

the direct self-similar Malý dipole mechanism cannot refute the AP-E12 endpoint.

(17)

This is a mechanism-specific no-go, not a proof that no more anisotropic or iterated concentration construction exists.

5 Finite energy does not imply continuity

The regularity fallback cannot begin by assuming that every graph-space map is continuous. Around a spatial axis set

$$u(r, z) = (\cos \ell(r), \sin \ell(r), 0, 0), \quad \ell(r) = \log \log(e/r). \quad (18)$$

Then Du has rank one, so $M_2(Du) = M_3(Du) = 0$, while

$$\int_0^{r_0} |\partial_r u|^2 r \, dr = \int_0^{r_0} \frac{dr}{r \log^2(e/r)} < \infty. \quad (19)$$

The phase has no limit as $r \downarrow 0$, so there is no continuous representative on the axis. Nevertheless, making ℓ constant below ϵ produces smooth rank-one truncations with zero higher minors and $W^{1,2}$ tail energy tending to zero. This example is therefore not a density counterexample. It proves only that results assuming continuous Cartesian maps cannot cover the full energy class without another argument.

6 The relaxed-minimizer fallback and its exact obstruction

Add the lower-order mass potential and define

$$E_B(u) = \int_{\Omega} \{W(Du) + m^2(1 - u^0)\} \, dx, \quad u \in \mathcal{G}_B^2, \quad (20)$$

and let $\overline{E}_B$ be its lower-semicontinuous relaxation from smooth fixed-degree maps. AP-E11 proves convergence to minimizers of $\overline{E}_B$, not equality $\overline{E}_B = E_B$ on all of $\mathcal{G}_B^2$.

For a classical map, put $F = Du$ and $G = F^T F$. Cauchy–Binet gives $|M_3(F)|^2 = \det G$ and $|M_2(F)|^2 = ((\operatorname{tr} G)^2 - \operatorname{tr} G^2)/2$. Hence the exact first Piola stress is

$$P(F) = DW(F) = F + R\{(\operatorname{tr} G)F - FG\} + KF \operatorname{cof} G. \quad (21)$$

The formal tangent Euler–Lagrange system is

$$(I - u \otimes u)\{-\operatorname{div} P(Du) - m^2 e_0\} = 0. \quad (22)$$

Proposition 6.1 (Dependency chain for the fallback). *A global minimizer of $\overline{E}_B$ may be promoted to a classical isolated solution of (22) only after the following independent steps are supplied:*

1. *local target-valued recovery with fixed trace, identifying relaxed compactly supported variations with variations of E_B ;*
2. *a weak Euler–Lagrange theorem for the resulting graph-space local minimizer;*
3. *an ϵ -regularity or partial-regularity theorem compatible with the $\mathbb{S}^3$ constraint and complete-minor energy;*
4. *removal of the possible singular set and a strict local growth estimate modulo translations and the global target $SO(3)$ orbit.*

Without item 1, minimality of the relaxation does not logically imply (22) for the naive integral.

The recent theorem of Gmeineder and Kristensen proves smooth partial regularity of relaxed minimizers for strongly quasiconvex (p, q) integrands in the range

$$1 \leq p \leq q < \min \left\{ \frac{np}{n-1}, p+1 \right\} [5]. \quad (23)$$

Here the pointwise bounds are $c|F|^2 \lesssim W(F) \lesssim C(1 + |F|^6)$, so $(n, p, q) = (3, 2, 6)$ and (23) would require $q < 3$. Schäffner’s higher-integrability theorem for convex nonstandard-growth integrands requires $q/p < 1 + 2/(n-1) = 2$, whereas the present ratio is three and W is polyconvex rather than convex in F [6]. Neither theorem includes the sphere constraint or fixed-degree graph relaxation. The exact identity (9) is promising additional structure, but applying it requires the missing trace-preserving graph competitor.

Consequently the requested fallback has been executed as far as presently justified and stops before its first promotion:

$$\boxed{C_{\text{local recovery}} = C_{\text{weak EL}} = C_{\text{partial regularity}} = C_{\text{classicality}} = C_{\text{isolation}} = \text{false}.} \quad (24)$$

This is an absence of proof, not a counterexample to regularity of the particular numerical $B = 1$ minimizer.

7 Numerical concentration and isolation proxy

The production script re-relaxes the same AP-E11 action at three spacings and volumes. No center or core value is pinned. The maximum fraction of total energy in one tetrahedron decreases under refinement, as does the 99.9th percentile of the local energy density:

N	L	a	E	max tet fraction	density $q_{0.999}$
17	5.5	0.343750	77.5780	3.966×10^{-3}	31.187
21	6.0	0.300000	79.2457	1.849×10^{-3}	24.900
25	6.5	0.270833	79.9500	1.173×10^{-3}	20.971

All three have $B = (1, 1, 1)$, minimum pair dot at least 0.2724, and projected gradient density at most 1.54×10^{-7} . In the smallest audited balls, whose radii are approximately two lattice spacings, the energy fractions decrease $0.420 \rightarrow 0.264 \rightarrow 0.189$. These observations exclude a visible one-cell concentration in this range; they are not a compensated-compactness or ϵ -regularity theorem.

Three independent smooth tangent perturbations of amplitude 0.24 are then re-relaxed on the $N = 17$ action. Field distance is evaluated after quotienting integer spatial translations and the unbroken target $SO(3)$. If the pion parts of reference and candidate are matrices X, Y , the proper Procrustes rotation is

$$Y^T X = U \Sigma V^T, \quad Q_* = U \operatorname{diag}(1, 1, \det(UV^T)) V^T \in SO(3). \quad (25)$$

The maximum results over the three starts are

$$\frac{\max E - \min E}{\bar{E}} = 3.38 \times 10^{-11}, \quad d_{\text{CDF}} = 5.70 \times 10^{-7}, \quad d_{\text{field}/(\text{trans} \times \text{SO}(3))} = 1.31 \times 10^{-5}. \quad (26)$$

All endpoints remain $B = (1, 1, 1)$ and stationary below 4.31×10^{-7} . The predeclared finite-dimensional basin proxy passes. It is deliberately not promoted to continuum isolation because it samples only three perturbations on one finite grid and does not give a quantitative neighbourhood theorem.

8 Decision and next theorem

The reproducible card

`route_f/code/verify_ap_e12_graph_density_regular_minimizer.py`

passes 9/9 algebraic, scaling, quadrature, concentration, and finite-basin checks. The logical authorization is nevertheless

$$\begin{aligned} C_{\text{degree closed in graph}} &= C_{\text{LH}} = C_{\text{strong QC}} = C_{\text{dipole no-go}} = \text{true}, \\ C_{D_{\text{end}}} &= \text{false (not disproved)}, \\ C_{\text{partial regularity}} &= C_{\text{classicality}} = C_{\text{isolation}} = C_{\text{Hessian}} = C_{\text{det}} = \text{false}. \end{aligned} \quad (27)$$

The most targeted next lemma is an annular, target-valued endpoint replacement: given a low-energy shell, choose a good sphere and replace its interior while preserving the trace and obtaining equiintegrability of all three L^2 minor families. Combined with (9), such a lemma would yield either a graph-density proof or the Caccioppoli inequality needed for the fallback. The second route is to prove (11) directly for the selected relaxed minimizer and then establish the sphere-valued projection/retraction step. Only after one route proves classicality and continuum isolation is it legal to assemble the same-action Riemann Hessian. The general danger that a relaxed and a smooth variational problem need not have the same infimum is precisely the Lavrentiev boundary emphasized in recent reviews [7]; it must not be removed by numerical agreement alone.

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